



Roland Berger AI Lab

Take-Home Assignment: Strategy Document Analysis Agent

Background

Build a simple generative AI agent that analyzes business and strategy documents and returns a structured assessment.

The purpose of this assignment is to evaluate your ability to research unfamiliar topics, make reasonable technical decisions, experiment, and communicate your reasoning. We are not expecting a production-ready application.

Use Case

The user uploads a strategy-related PDF or Word document, such as:

- A corporate strategy document
- A market-entry study
- A business plan
- A transformation roadmap
- A market research report

The agent must analyze the document and produce a structured consulting-style review.

Functional Requirements

The solution should:

1. Accept PDF and DOCX files (you can create synthetic data to test on it or use any relevant online docs).
2. Extract and process the document's textual content.
3. Use a clearly defined system prompt.
4. Produce output that conforms to a defined JSON schema.
5. Validate the generated output before returning it.
6. Include references to relevant pages or sections where possible.
7. Handle documents that contain images, diagrams, tables, or charts.
8. Communicate errors, uncertainty, or unsupported content clearly.

Expected Structured Output

The output should contain at least:

- Document title
- Document type
- Executive summary
- Strategic objective
- Key findings
- Main recommendations
- Important business metrics
- Assumptions
- Risks or concerns
- Missing or unclear information
- Recommended follow-up questions
- Evidence references and confidence levels

You may modify or extend the schema if you explain your reasoning.

Visual Content Challenge

Some documents may include important information inside:

- Market-sizing charts
- Financial tables
- Competitor matrices
- Transformation roadmaps
- Operating-model diagrams
- Scanned pages
- Embedded images
- Tables that are not extracted correctly by normal parsers

Explain and demonstrate how your solution attempts to process this content.

It is acceptable if the solution has limitations. Clearly document what works, what does not work, and what you would improve with more time.

Deliverables

Please provide:

1. Source code with instructions for running it.
2. The system prompt and structured-output schema.
3. A short README describing the architecture.
4. At least two example documents or test cases.
5. Example outputs.
6. A brief explanation of:
 - Libraries and models considered
 - Decisions made
 - Experiments attempted
 - Problems encountered
 - Known limitations
 - Improvements you would make next
7. A short demonstration, either live or recorded.

Evaluation Criteria

We will evaluate:

- Problem decomposition
- Independent research
- Critical thinking
- Quality of experimentation
- Prompt and schema design
- Reliability of structured output
- Ability to distinguish evidence from assumptions
- Handling of visual document content
- Error handling
- Awareness of limitations

Helpful Resources and Guidelines

- You may use any programming language, libraries, frameworks, AI tools, or documentation.
- For access to an AI model, you may create an OpenRouter API key and use any available free model.
- Free models may have limited availability, lower rate limits, or varying support for vision and structured output, so **document any constraints you encounter**.
- You may use AI coding assistants, but you should understand and be able to explain the submitted solution.